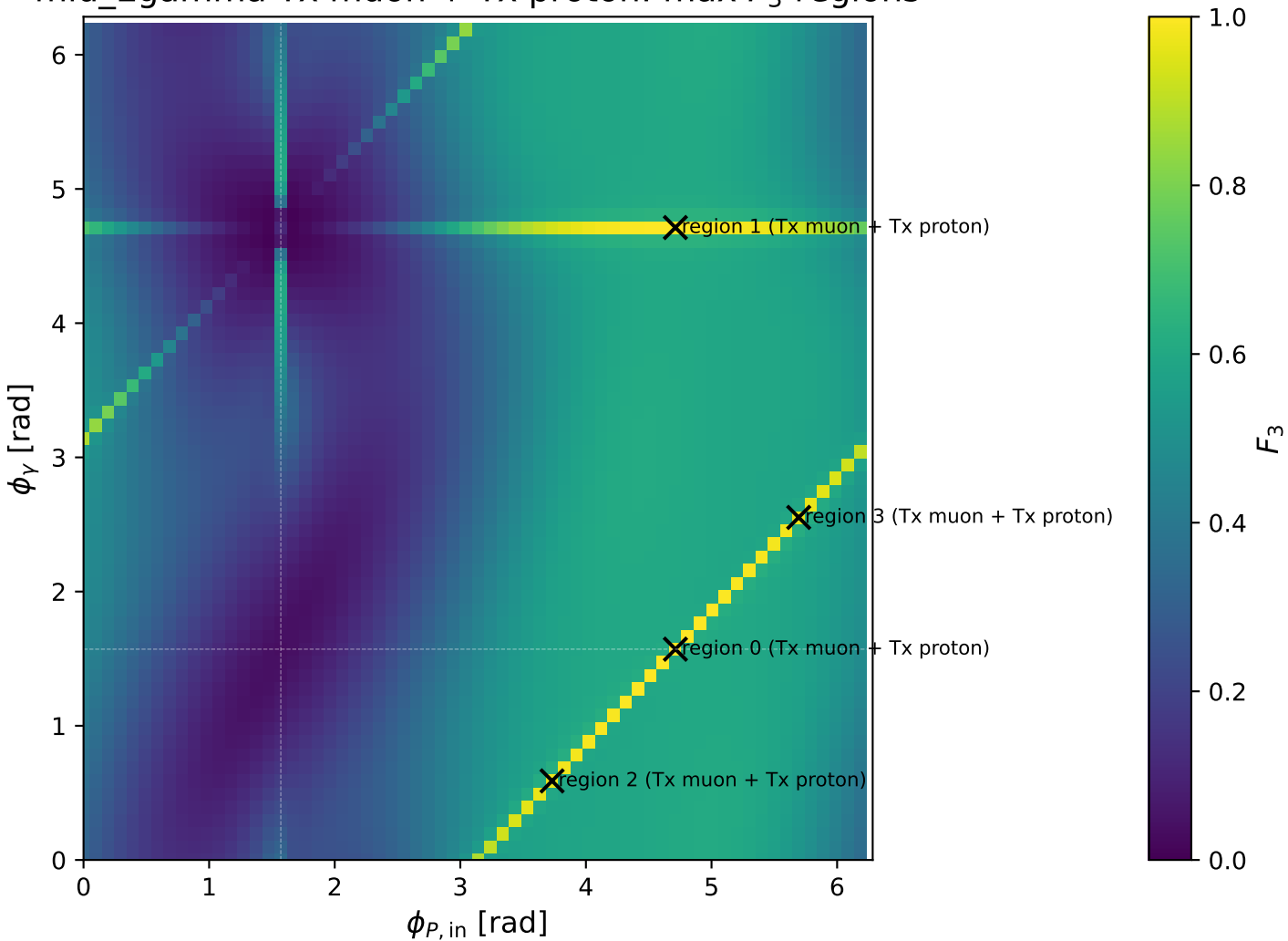
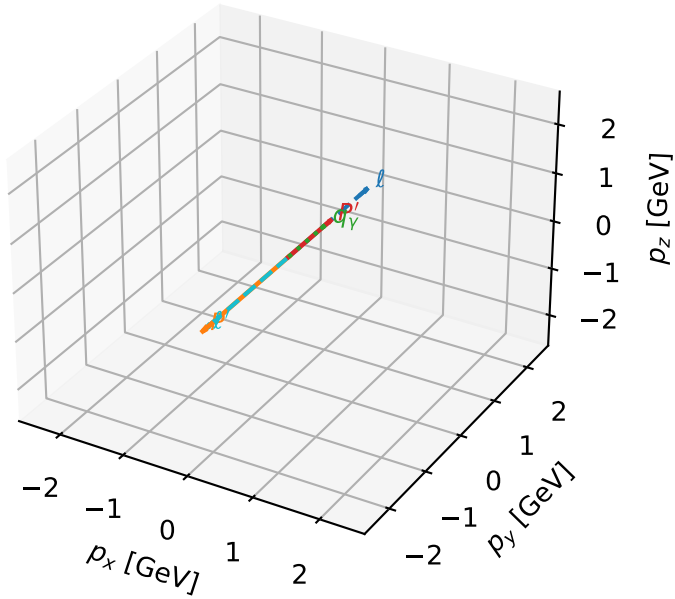


mid_Egamma Tx muon + Tx proton: max F_3 regions



Tx muon + Tx proton characteristic max F_3 configuration

3D momenta

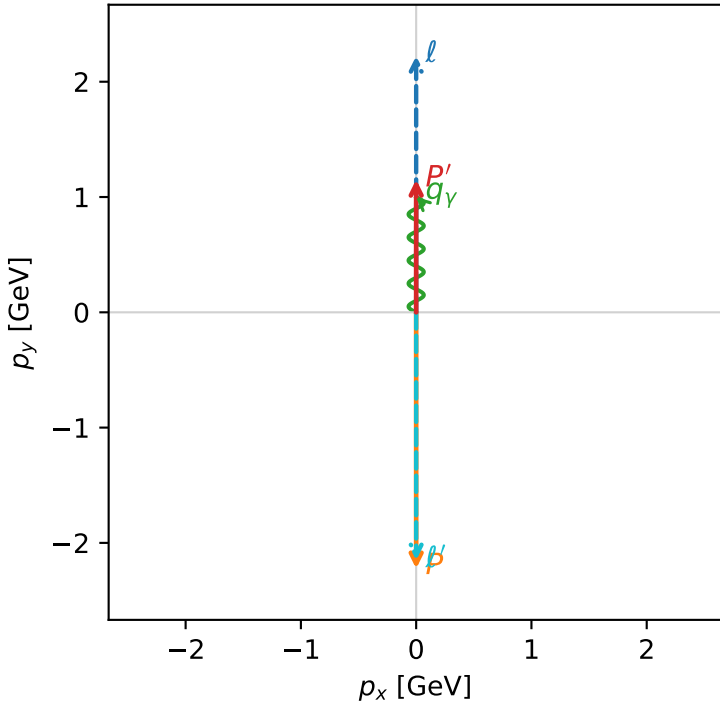


f3_mid_egamma_txtx_region_0 F_3=1
region: mid_Egamma_TxTx_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.15107$
 $\theta_{in}=1.57079$, $\phi_l=1.5708$, $\phi_p=4.71239$
 $E_\gamma=1$, $\phi_\gamma=1.5708$
 $Q^2=19.1283$, $x_B=0.966277$, $t=-10.5238$, $W^2=1.54742$, $y=0.959806$
 $\theta(l', \gamma)=180$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= 0.0000$
 l' $E= 2.1537$ $p_x= -0.0000$ $p_y= -2.1511$ $p_z= 0.0000$
 P' $E= 1.4849$ $p_x= 0.0000$ $p_y= 1.1511$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.0000$ $p_y= 1.0000$ $p_z= 0.0000$

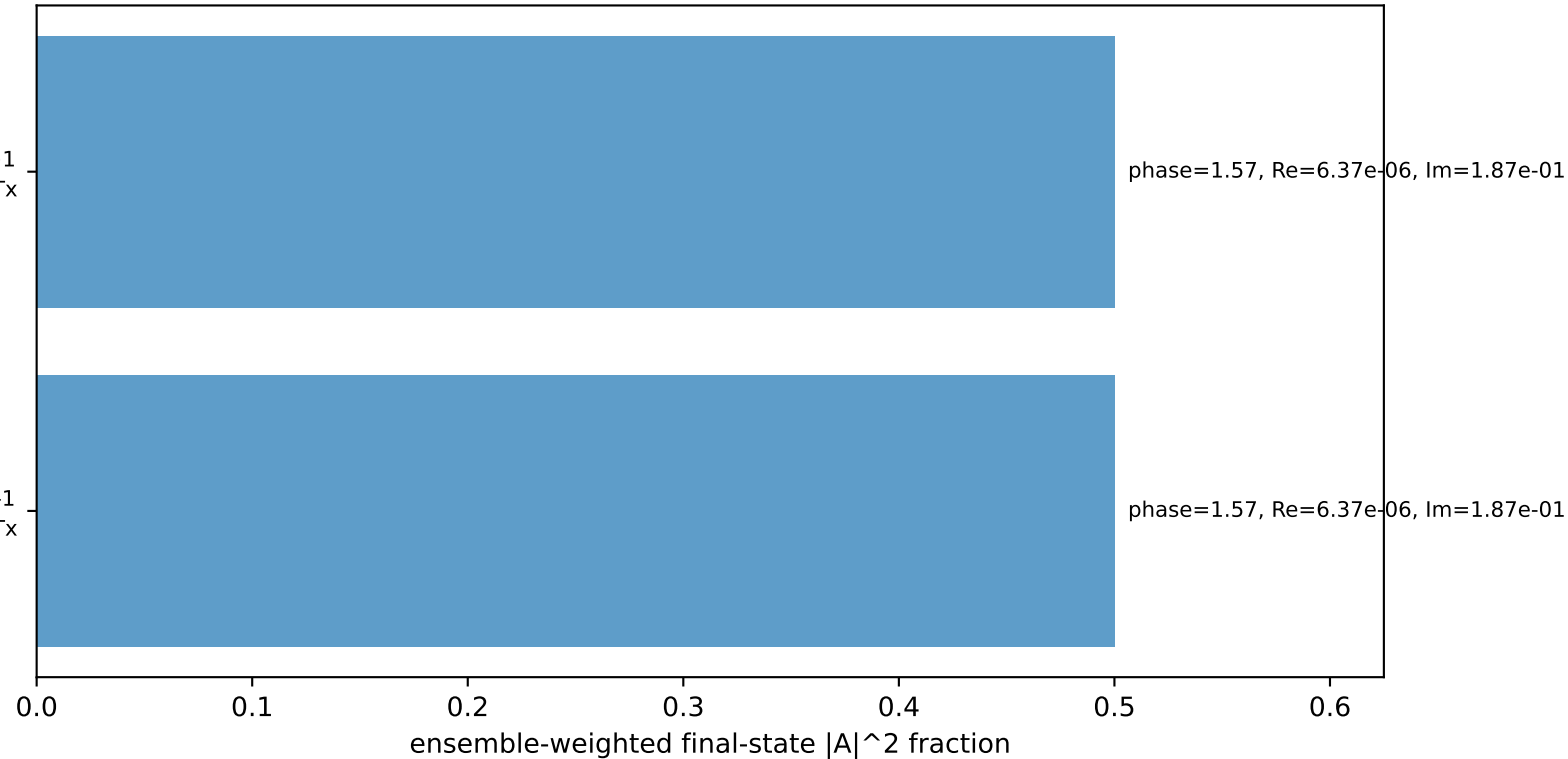
Transverse plane



$h_e=-1$, $h_p=-1$, $h_\gamma=+1$
electron Tx, proton Tx

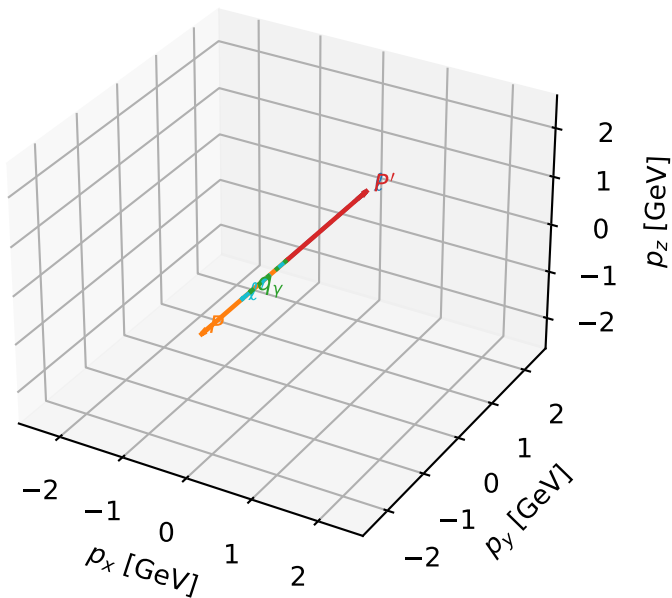
$h_e=+1$, $h_p=+1$, $h_\gamma=-1$
electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



Tx muon + Tx proton characteristic max F_3 configuration

3D momenta

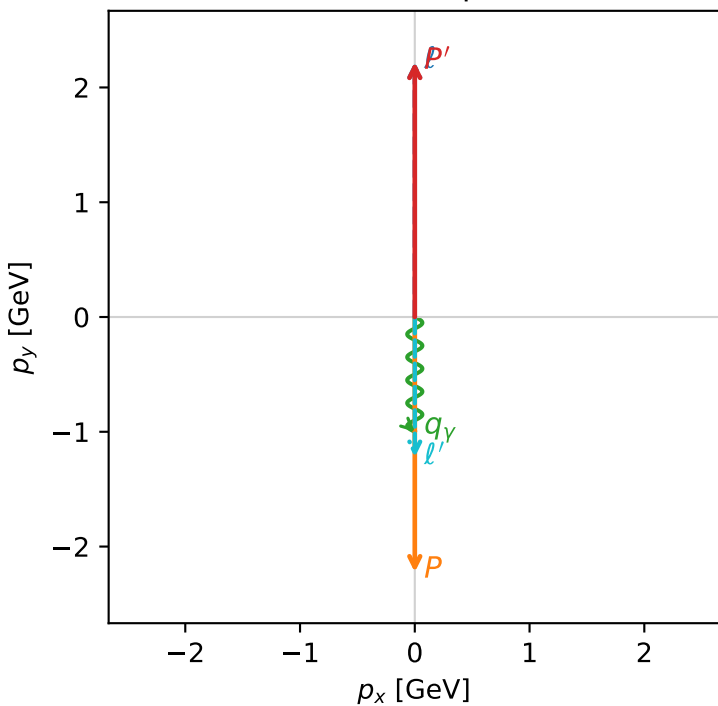


f3_mid_egamma_txtx_region_1 F_3=0.999999
 region: mid_Egamma_TxTx_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |\bar{T}_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.22204$
 $\theta_{in}=1.57079$, $\phi_l=1.5708$, $\phi_p=4.71239$
 $E_Y=1$, $\phi_Y=4.71239$
 $Q^2=10.8711$, $x_B=0.539802$, $t=-19.7594$, $W^2=10.1478$, $y=0.976443$
 $\theta(l',\gamma)=0$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= 0.0000$
 l' $E= 1.2266$ $p_x= 0.0000$ $p_y= -1.2220$ $p_z= 0.0000$
 P' $E= 2.4119$ $p_x= 0.0000$ $p_y= 2.2220$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.0000$ $p_y= -1.0000$ $p_z= 0.0000$

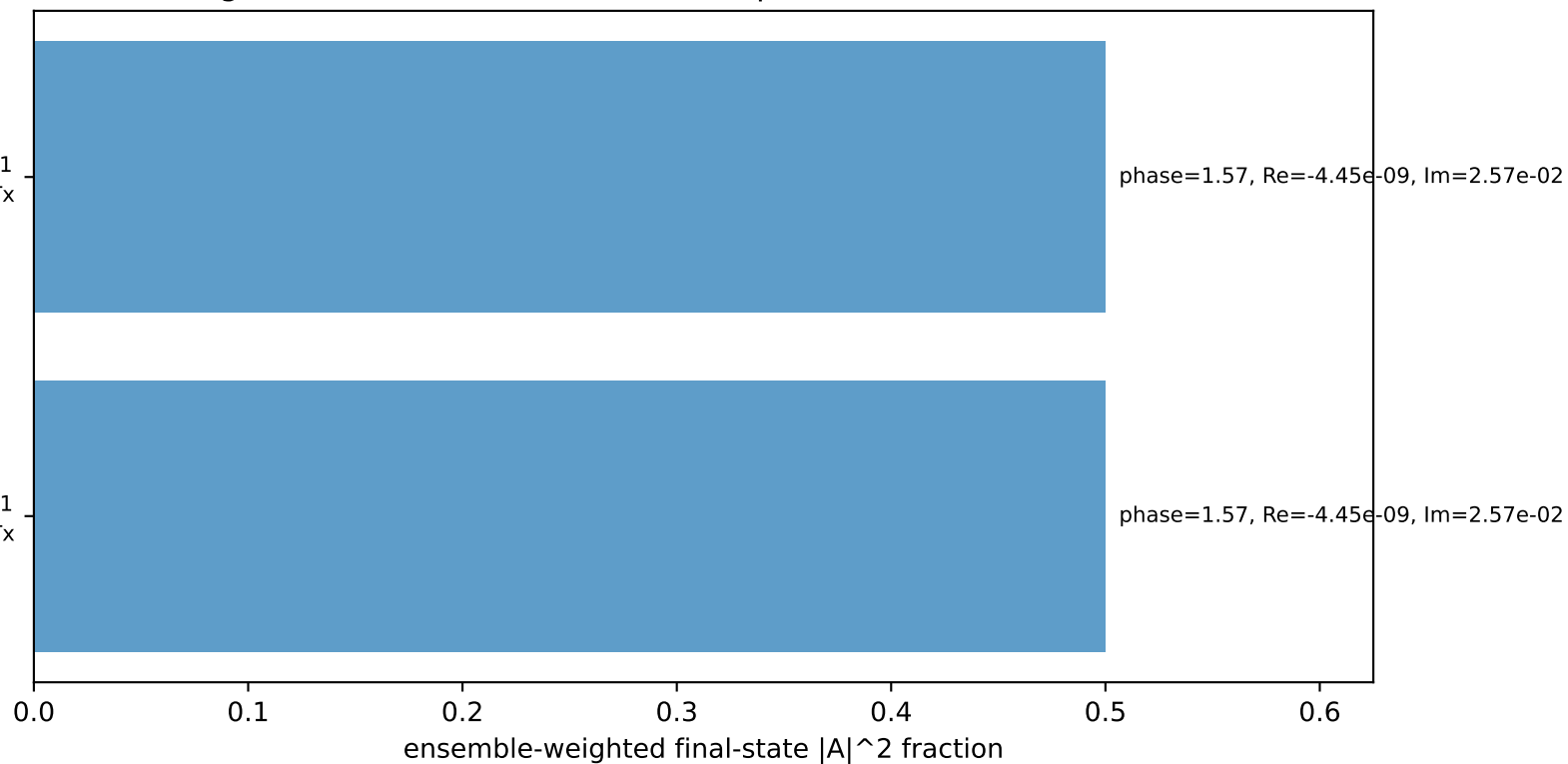
Transverse plane



$h_e=-1$, $h_p=+1$, $h_Y=+1$
 electron Tx, proton Tx

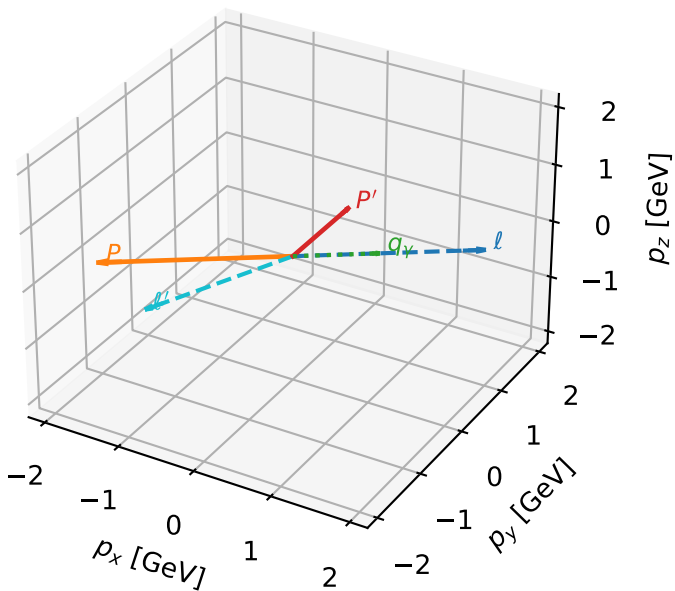
$h_e=+1$, $h_p=-1$, $h_Y=-1$
 electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



Tx muon + Tx proton characteristic max F_3 configuration

3D momenta

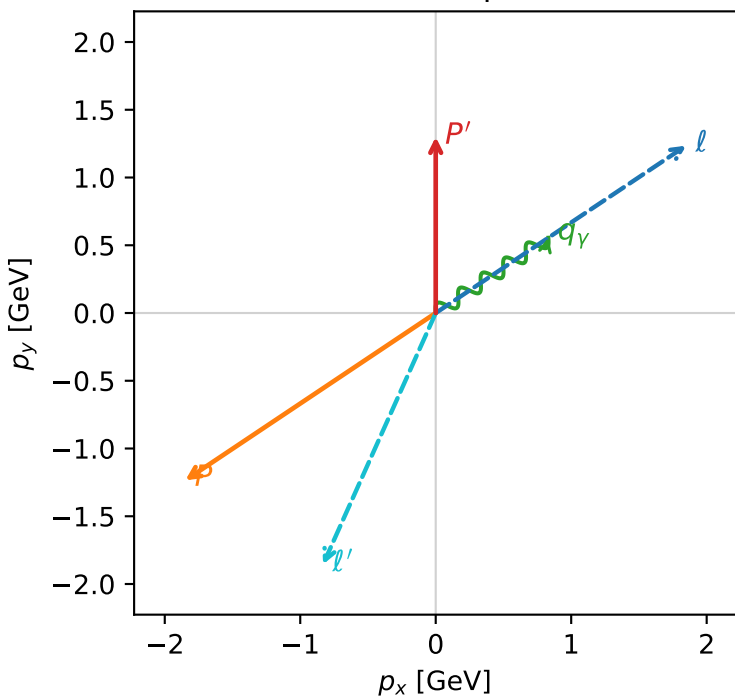


f3_mid_egamma_txtx_region_2 $F_3=0.971496$
 region: mid_Egamma_TxTx_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

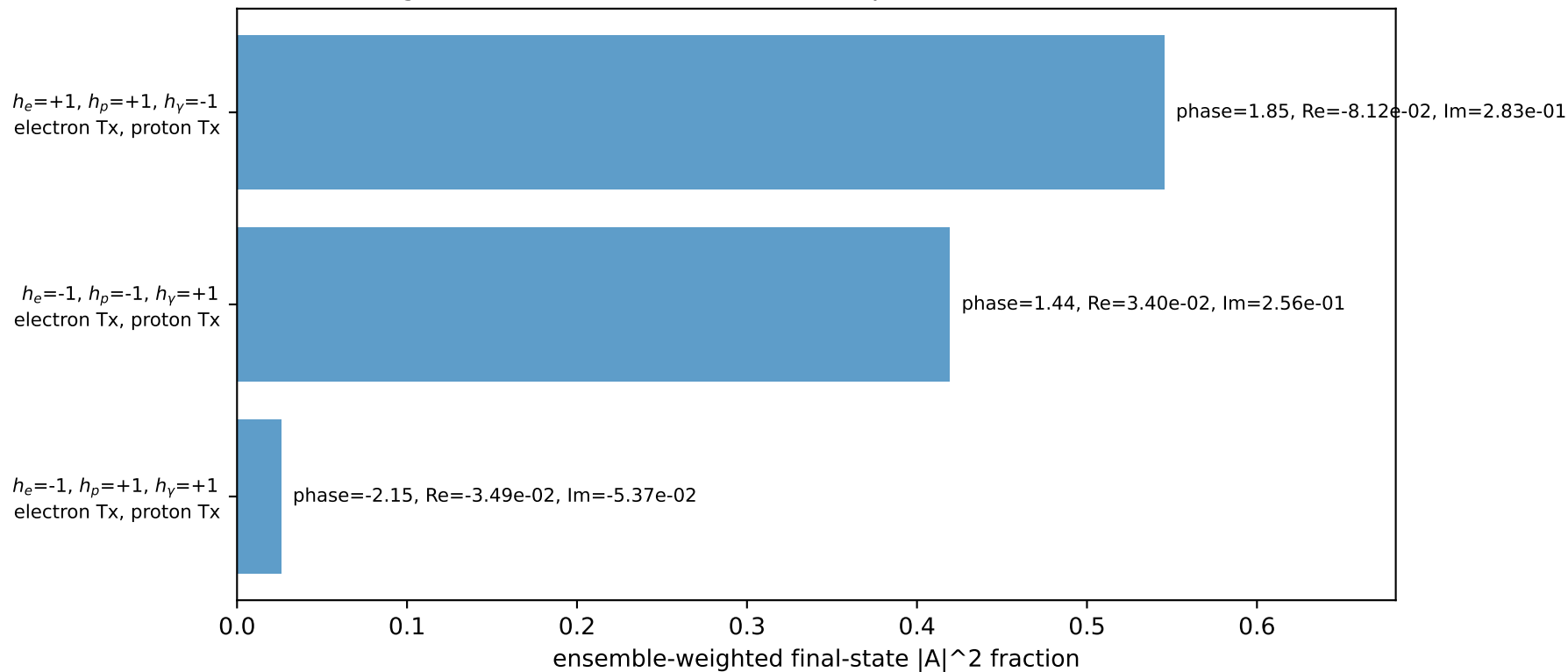
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.29963$
 $\theta_{in}=1.57079$, $\phi_l=0.589049$, $\phi_p=3.73064$
 $E_Y=1$, $\phi_Y=0.589049$
 $Q^2=16.6958$, $x_B=0.904565$, $t=-9.18528$, $W^2=2.64132$, $y=0.894906$
 $\theta(l', \gamma)=147.891$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 1.8484$ $p_y= 1.2351$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.8484$ $p_y= -1.2351$ $p_z= 0.0000$
 l' $E= 2.0357$ $p_x= -0.8315$ $p_y= -1.8552$ $p_z= 0.0000$
 P' $E= 1.6028$ $p_x= 0.0000$ $p_y= 1.2996$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.8315$ $p_y= 0.5556$ $p_z= 0.0000$

Transverse plane

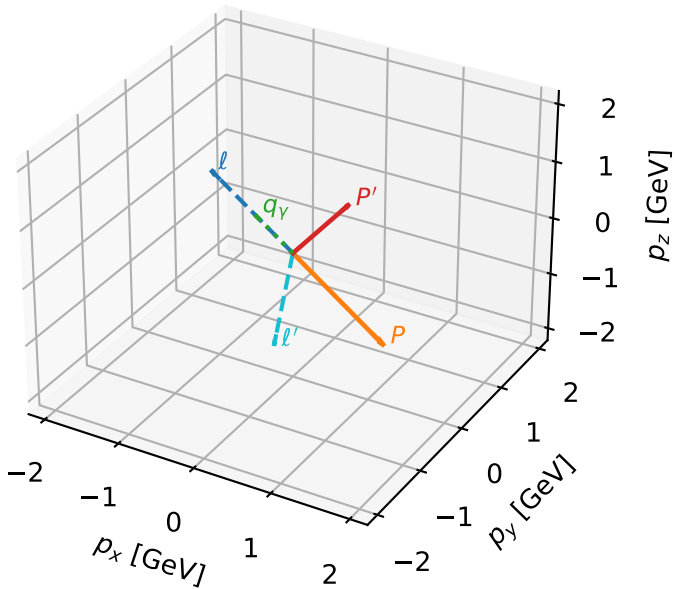


Leading initial-ensemble/final-state components (N=3, retained=99.0%)



Tx muon + Tx proton characteristic max F_3 configuration

3D momenta

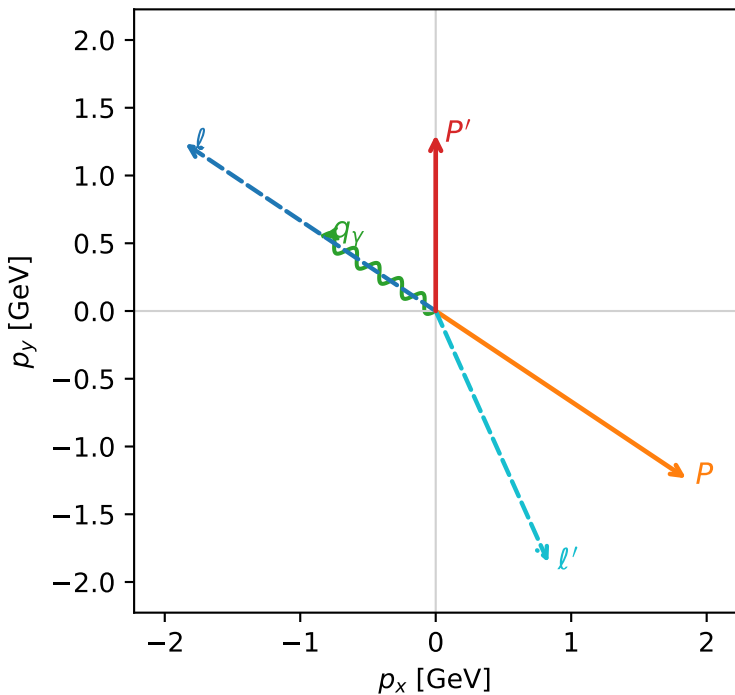


f3_mid_egamma_txtx_region_3 $F_3=0.971496$
 region: mid_Egamma_TxTx_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.29963$
 $\theta_{in}=1.57079$, $\phi_l=2.55254$, $\phi_P=5.69414$
 $E_\gamma=1$, $\phi_\gamma=2.55254$
 $Q^2=16.6958$, $x_B=0.904565$, $t=-9.18528$, $W^2=2.64132$, $y=0.894906$
 $\theta(l', \gamma)=147.891$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -1.8484$ $p_y= 1.2351$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.8484$ $p_y= -1.2351$ $p_z= 0.0000$
 l' $E= 2.0357$ $p_x= 0.8315$ $p_y= -1.8552$ $p_z= 0.0000$
 P' $E= 1.6028$ $p_x= 0.0000$ $p_y= 1.2996$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.8315$ $p_y= 0.5556$ $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
 electron Tx, proton Tx

$h_e=+1, h_p=+1, h_\gamma=-1$
 electron Tx, proton Tx

$h_e=+1, h_p=-1, h_\gamma=-1$
 electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=3, retained=99.0%)

